

Week 3: Stochastic Optimization

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What stood out to me in Hardt and Recht's optimization chapter was how much the choice of objective matters for fitting a model. Empirical risk minimization means minimizing average training loss, which connects directly to last week's least-squares and likelihood objectives. Convexity helps explain why some of these problems are tractable: for a differentiable convex objective, any stationary point is a global minimum. Still, convergence of gradient descent requires appropriate conditions, including the choice of step size. I found this distinction useful because it separates properties of the problem from choices we make when solving it.

I also found the tradeoff behind stochastic gradient descent (SGD) interesting. For the average loss $F(w) = n^{-1} \sum_{i=1}^n \ell_i(w)$, sampling an example uniformly gives a gradient whose expectation is $\nabla F(w)$. Each update is much cheaper than computing the full gradient, although an individual step can increase the training loss. This makes SGD's usefulness easier to understand: we can afford many imperfect updates. Near an optimum, individual examples can still push the parameters in different directions even when their gradients average to zero. The learning-rate schedule therefore affects both how quickly we make progress and how much we continue to fluctuate.

Gower et al.'s discussion of variance reduction gives a useful response to this problem. For example, SVRG periodically computes a full gradient at a reference point and uses it to correct subsequent stochastic estimates. I like the idea of spending additional computation occasionally to make many later updates more informative. However, Bottou et al.'s distinction between optimization and statistical error makes me hesitate to equate a more accurate training solution with a better model. Improving predictions also depends on the data and the model class. The question I am left with is how, under a fixed computing budget, we should decide whether to spend more effort reducing gradient noise or whether ordinary SGD has already optimized the training objective well enough.

Readings: Hardt and Recht, "Optimization," Chapter 5; Gower et al., *Variance-Reduced Methods for Machine Learning*; Bottou et al., *Optimization Methods for Large-Scale Machine Learning*.